STAT 5243 Project 1: WallStreetBets Data Pipeline and Exploratory Analytics

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February 19, 2026

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1 Introduction and Dataset Description

1.1 Introduction

The rise of online social platforms has reshaped modern financial markets by expanding access to information and collective participation. The Reddit community `r/WallStreetBets` (WSB) became a major retail-investor forum during 2020–2021, especially throughout the meme-stock period. This project analyzes WSB post activity to study how online attention is reflected in engagement behavior.

Our objective is to transform raw social-media data into a reliable analytical dataset and derive interpretable insights about investor discussion patterns. The analysis is designed as a complete data-science workflow, from acquisition through preprocessing, EDA, feature engineering, and report synthesis.

1.2 Dataset Description

The dataset contains 53,187 WSB posts, where each row corresponds to one Reddit submission. Core variables include `title`, `score`, `id`, `url`, `comms_num`, `created`, `body`, and `timestamp`. The dataset combines numerical engagement metrics, unstructured text, and temporal metadata.

The raw data has notable real-world complexity:

- Unstructured text with markdown, URLs, and internet slang.
- High missingness in `body`, largely structural by post type.
- Heavy-tailed engagement metrics with extreme viral outliers.
- Mixed temporal representations requiring consistency checks.

These properties make the dataset suitable for advanced cleaning, preprocessing, and exploratory analysis.

2 Data Acquisition Methodology

The project uses the WallStreetBets Reddit post dataset distributed via Kaggle as the primary public repository source. The data was originally collected from `https://www.reddit.com/r/wallstreetbets/` using PRAW (Python Reddit API Wrapper), then shared as a CSV file for research and educational use.

2.1 Acquisition Procedure

1. Select an open-access dataset relevant to retail-investor discussion behavior.
2. Download raw CSV data from Kaggle and preserve it as the immutable source file.
3. Validate schema, column semantics, and temporal coverage before transformation.
4. Import raw data into Python environment for cleaning and preprocessing.

2.2 Scope and Relevance

The observation window (2020–2021) captures the meme-stock period, making it suitable for studying rapid shifts in community attention and engagement. The dataset includes both structured metadata and unstructured text, which supports a full preprocessing and feature-engineering workflow.

2.3 Acquisition Constraints

The dataset reflects available post-level metadata and does not include direct trade execution records. As a result, downstream analysis focuses on discussion and engagement proxies rather than verified transaction behavior.

3 Cleaning and Preprocessing Steps

The cleaning workflow standardizes data types, removes formatting artifacts, evaluates duplicates, handles missingness, and stabilizes heavy-tailed engagement variables using log transforms. Additional preprocessing steps include scaling (z-score and min-max), categorical encoding, and feature derivation for downstream analysis.

Primary implementation is documented in `Project1.ipynb` and reproduced in script form in `code/cleaning_workflow.py`.

4 Exploratory Data Analysis (EDA)

EDA quantified engagement distribution shape, post-type composition, and temporal dynamics. The analysis includes summary statistics, distributional visualizations, relationship plots, and trend diagnostics across time slices.

Advanced diagnostics include correlation analysis, non-parametric group comparisons, and anomaly-rate summaries to strengthen interpretation beyond descriptive plotting.

5 Feature Engineering Process and Justification

Feature engineering extends baseline metadata with behavioral and content-aware variables to improve analytical usability and predictive signal quality.

5.1 Constructed Feature Families

- Temporal and engagement-derived features (existing branch baseline).
- Sentiment proxy from title text using a finance-oriented rule lexicon.
- Topic features from LDA over title text to capture latent discussion themes.
- Interaction-ready feature matrix used for viral-classification diagnostics.

5.2 Justification

These additions improve interpretability and predictive structure by incorporating both textual semantics and engagement behavior. Diagnostic plots (ROC, precision-recall, confusion matrix, calibration, and coefficient importance) provide transparent validation of feature usefulness.

6 Summary of Key Findings

6.1 Dataset-Level Findings

The cleaned analytical dataset contains 53,187 posts and 30 variables. A large share of posts have missing body text, indicating that discussion on WSB is frequently title-driven or media-linked. Engagement metrics are strongly right-skewed, with a small number of viral posts accounting for a large share of total attention.

6.2 EDA Findings

The EDA results show that:

- Raw `score` and `comms_num` distributions are highly heavy-tailed.
- Log-transformed engagement variables are more analytically stable and interpretable.
- Engagement is positively associated across voting and comment activity.
- Temporal effects are visible by posting hour, day-of-week, and event periods.
- Post type composition is dominated by text and image formats.

6.3 Interpretive Summary

The evidence suggests that retail attention in WSB is concentrated around high-visibility posts and event-driven spikes. Most posts receive limited interaction, while a small subset drives disproportionate community engagement, consistent with viral attention dynamics in online financial discourse.

7 Challenges Faced and Future Recommendations

7.1 Challenges Faced

Several technical and analytical challenges were encountered in this project:

- Post-level engagement metrics do not directly measure real investment execution or financial outcomes.
- Text content is unstructured and noisy, requiring substantial preprocessing before advanced NLP.
- Engagement variables are heavily skewed, making naive summary statistics sensitive to extreme viral posts.
- The 2020–2021 window reflects an unusual market regime, limiting generalization to other periods.

7.2 Future Recommendations

Future extensions that would improve explanatory power include:

- Advanced NLP (topic modeling, sentiment, NER) for richer narrative extraction.
- Integration with external market indicators (price, volume, volatility) for cross-domain linkage.
- Temporal modeling of attention propagation around major events.
- Cross-community comparison beyond WSB to evaluate transferability of findings.

These extensions would strengthen causal interpretation and improve the generalizability of investor-attention analysis.

8 GitHub Repository Link

Public repository: `https://github.com/ZemingLiang/STAT-5243-Project-1-Team-2`

9 Each Member's Contribution

- **Zeming Liang**: Data cleaning and preprocessing; creation of `reddit_wsb.csv`, `reddit_wsb_cleaned.csv`, and `Project1.ipynb`; final repository organization and integration of PNG/JSON/output artifacts.
- **Zuer Weng**: Exploratory Data Analysis (EDA) implementation and EDA branch outputs.
- **Duoli Chen**: Report-writing narrative sections (introduction/acquisition/summary/challenges text files).
- **Isaac Beers**: Feature engineering notebook content, including viral-classification modeling and validation in the feature engineering workflow.

10 Representative Figures

11 Deliverables Included in This Branch

- Raw and cleaned datasets: `reddit_wsb.csv`, `reddit_wsb_cleaned.csv`
- Full workflow code files under `code/`
- Section report sources under `report/sections/`
- Per-figure PNG artifacts under `artifacts/figures/`
- Per-figure and per-section JSON artifacts under `artifacts/json/`

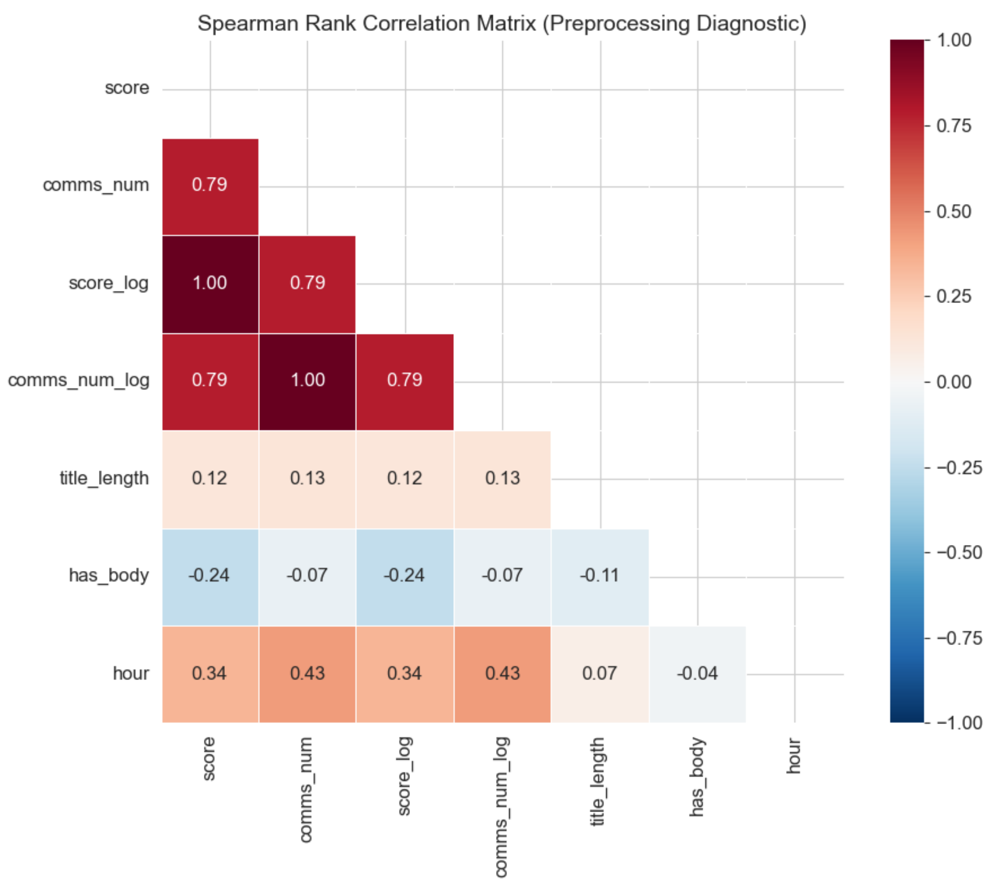

Figure 1: Cleaning/preprocessing diagnostic: Spearman correlation heatmap.

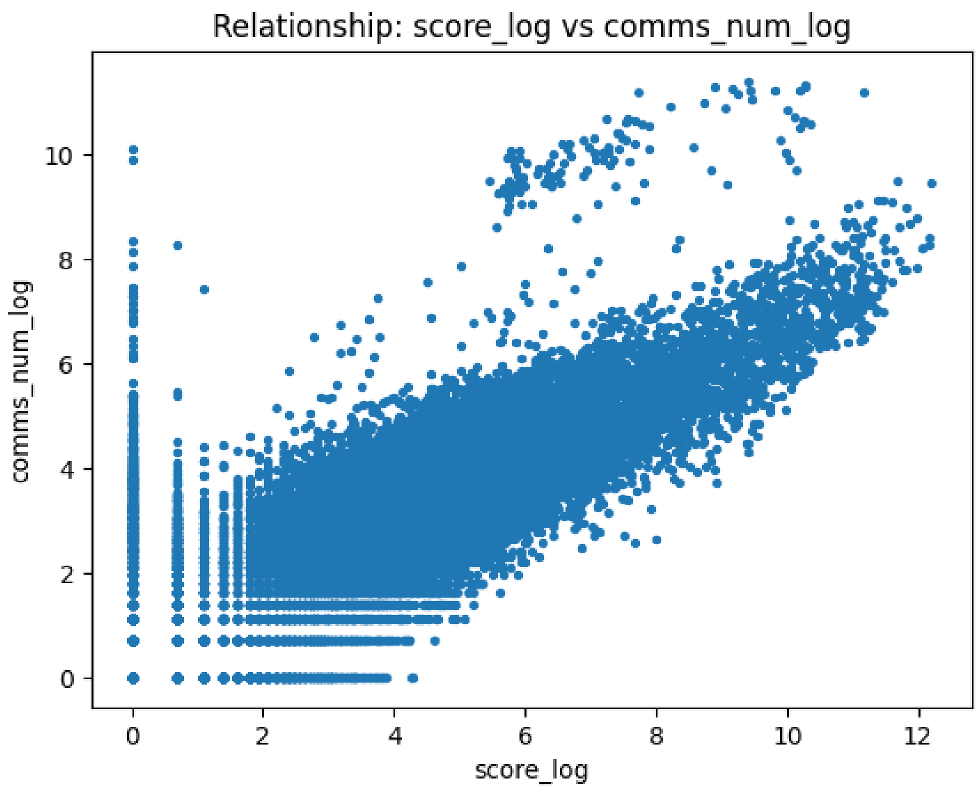

Figure 2: EDA relationship: `score_log` versus `comms_num_log`.

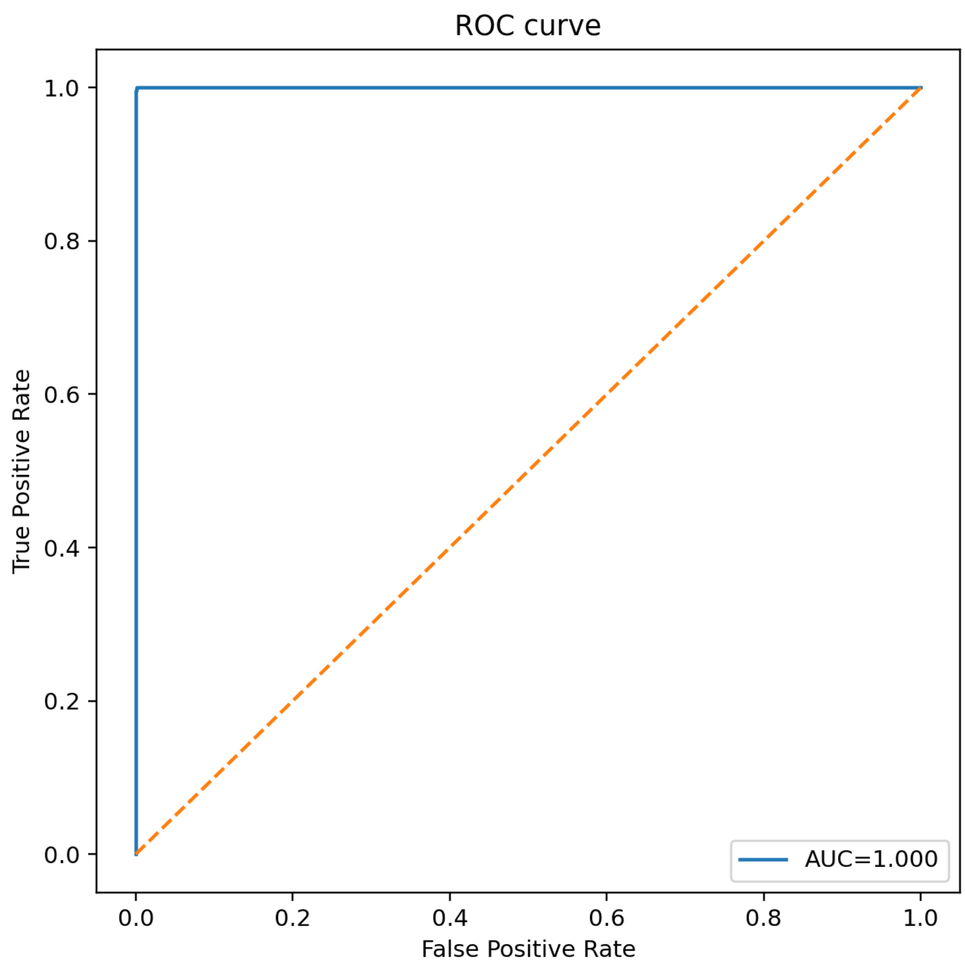

Figure 3: Feature engineering diagnostic: ROC curve for viral classification.